

ECEN 4517 - PV Power Lab
Experiment 3 Part 1 Report
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1 Buck Converter Design

1.1 Inductor Design

c) $L = 24.2 \mu\text{H}$ $K_n = 0.5$ Assumed

$$K_g \geq \frac{P L^2 I_{\text{max}}^2}{B_{\text{max}}^2 R K_n} 10^8 (\text{cm}^5) \quad K_g = \frac{A_c^2 W_A}{M L T}$$

Core Types: PQ 26/25 (N95) PQ 32/20 (N87)

$P_{\text{copper @ room temp: } 1.724 \times 10^{-4} \text{ } \Omega/\text{cm}$ Copper loss = $(85.14 \times 100\%)$
Copper loss = 0.426

N95 seems to have lower core losses @ 100 kHz compared to N87 $R = \frac{0.426}{6.55^2} = 9.92 \text{ m}\Omega$

N95 - $B_{\text{max}} = 525 \text{ mT}$ choose 250 mT

$$K_g \geq \frac{(1.724 \times 10^{-4} \times 24.2 \mu\text{H})^2 (7.35)^2}{(0.250)^2 (0.00992) (0.5)} 10^8 (\text{cm}^5)$$

$$K_g \geq 0.0176 \text{ cm}^5$$

PQ 26/25 $K_g = 0.125 \geq 0.00399$

$A_c = 1.18$
 $W_A = 0.503$
 $M L T = 5.62$
 $k_m = 5.55$
 $K_{gpc} = 9.4 \times 10^{-3}$

$$n = \frac{L I_{\text{max}}}{B_{\text{max}} A_c} 10^4 = \frac{(24.2 \mu\text{H}) (7.35)}{(0.250 \times 1.18)} 10^4$$

$n = 6 \text{ turns}$

c) Continued

$$\text{Airgap} = \frac{\mu_0 A_c n^2}{L} 10^{-4} = \frac{(4\pi \times 10^{-7}) (1.18) (6^2)}{24.2 \times 10^{-6}} 10^{-4} =$$

$l_g = 0.22 \text{ mm}$

Wire Gauge $A_w \leq \frac{K_u W_A}{n} (\text{cm}^2)$ AWG 12

$$\frac{(0.5 \times 0.503)}{6} = 0.0419 \text{ cm}^2 \geq A_w$$

$$41.9 \times 10^{-3} \text{ cm}^2 \geq 33.08 \times 10^{-3} \text{ cm}^2$$

Figure 1.1 - 1: Inductor Design Calculations

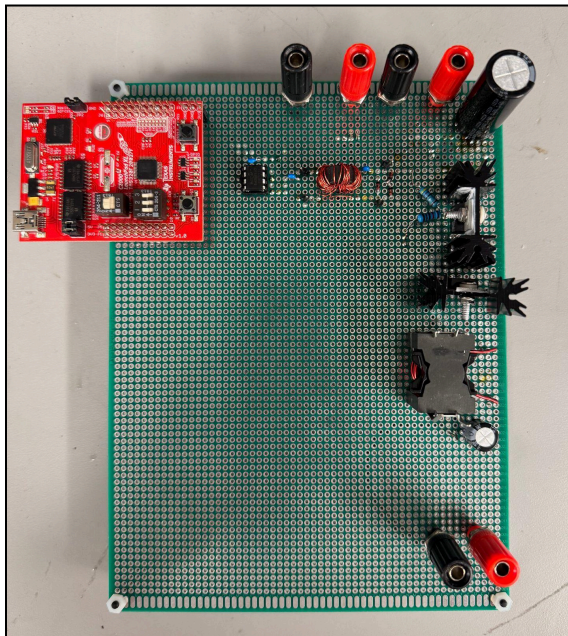
Above are the calculations used to design the inductor that would be used in the buck converter. It was designed for 100kHz, with 24.2 μH of inductance, and to use the PQ 26/25, N95 Ferrite Core. With this information, the number of turns required was calculated to be 6, with a minimum wire gauge of 12 AWG.

	Cursor 1
Frequency	100 kHz
Trace 1	Ls
Measurement	21.594 μ H

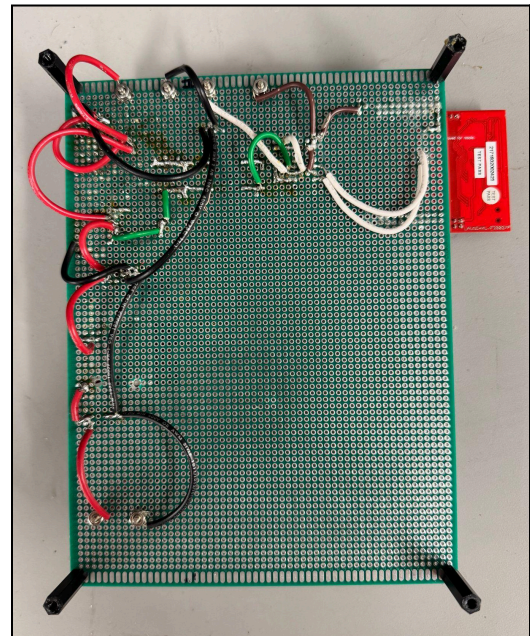
Figure 1.1 - 2: Inductor Measurement

Above is our measured inductance which is close to our calculated inductance shown in Figure 1.1 - 1. For our design we used an air gap of 0.0075 inches, with 6 turns, and a wire gauge of 18. We used a DC Resistance meter to measure R_s , which was 8.8 milliohms.

1.2 Buck Power Stage Construction



Top View



Bottom View

2 Buck Converter Testing

2.1 Buck Power Stage Testing

The goal of this testing was to test the buck converter at a load resistance modeling the lead acid battery we will be charging in a safe way. The desired output power was 85W with an output voltage of 13V. This would correspond to an output current of 6.54A. During the testing, a duty cycle of 80% was used in order to obtain the desired output. Ultimately, the measured output power of 85.8 W at 13V with 6.6A of current was taken with a load resistance of 2.4Ω . The input power was measured at 17.2V at 5.5A corresponding to 94.6W. This gives the converter an efficiency of 91%.

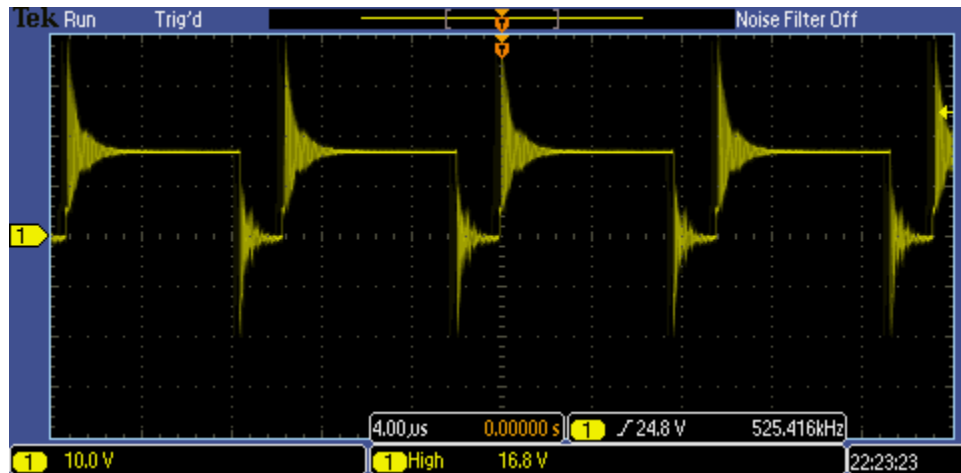


Figure 2.1 - 1: Diode Voltage - 10V and 4us per division

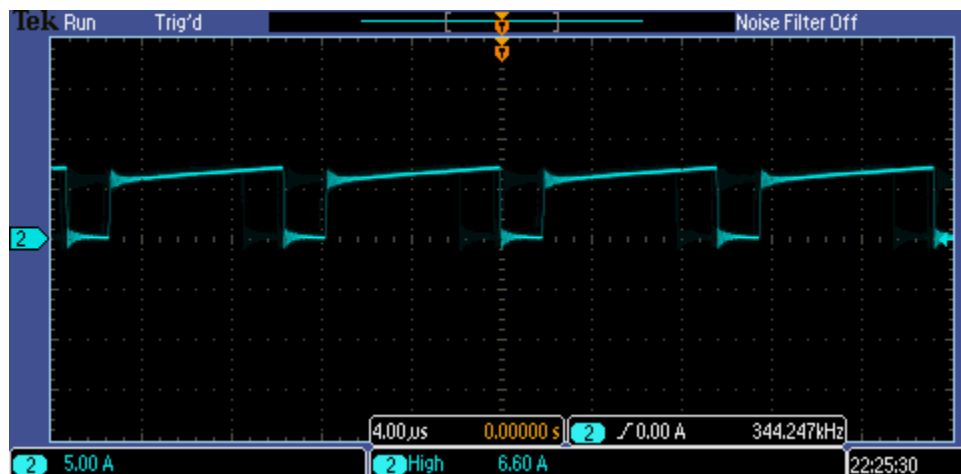


Figure 2.1 - 2: Transistor Current - 5A and 4us per division

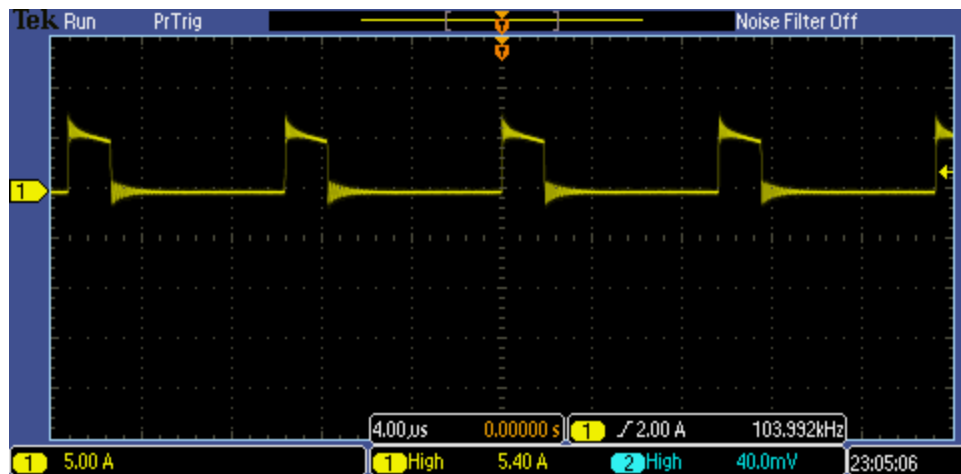


Figure 2.1 - 3: Diode Current - 5A and 4 μ s per division

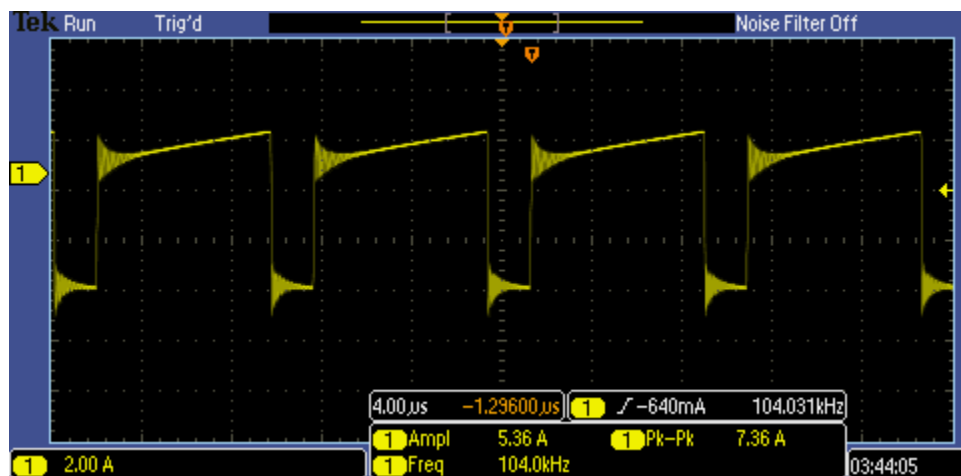


Figure 2.1 - 5: Capacitor C1 Current -2A and 4 μ s per division



Figure 2.1 - 6: Capacitor C1 Voltage - 10V and 4 μ s per division

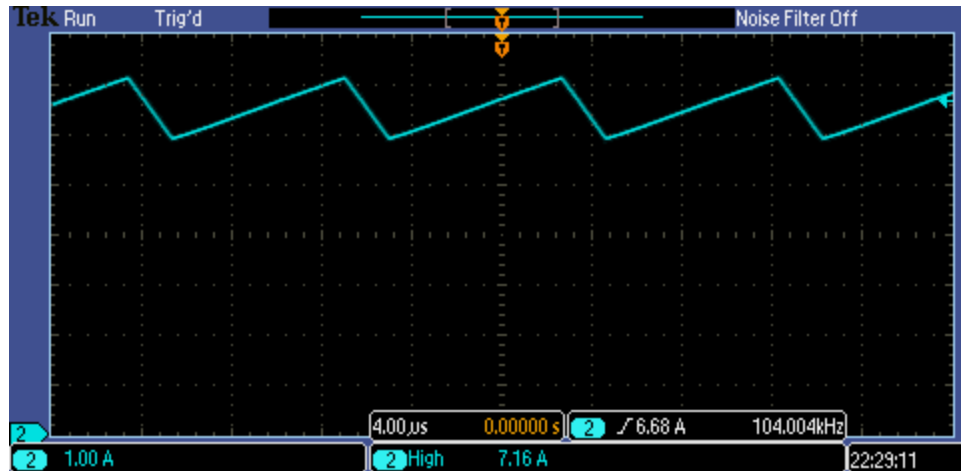


Figure 2.1 - 4: Inductor Current - 1A and 4us per division

Above is the Inductor current waveform taken during our testing. It measured a max inductor current of 7.16A with a mean value 6.68A. This would correspond to an inductor ripple current of 960mA. During the prelab of this experiment, an average inductor current and current ripple were calculated to help design the inductor overall. An average inductor current calculated to be 6.55A with a 20% ripple of 1.31A. The experimentally gathered values shows a 20% increase in average inductor current and a 26% decrease in inductor ripple. While these numbers may seem larger than expected, the overall result of the inductor design is a success. The buck converter is able to regulate its output voltage with respect to duty cycle and produce favorable results in all aspects of testing.

2.1.1 ESR Measurements of Capacitor C1

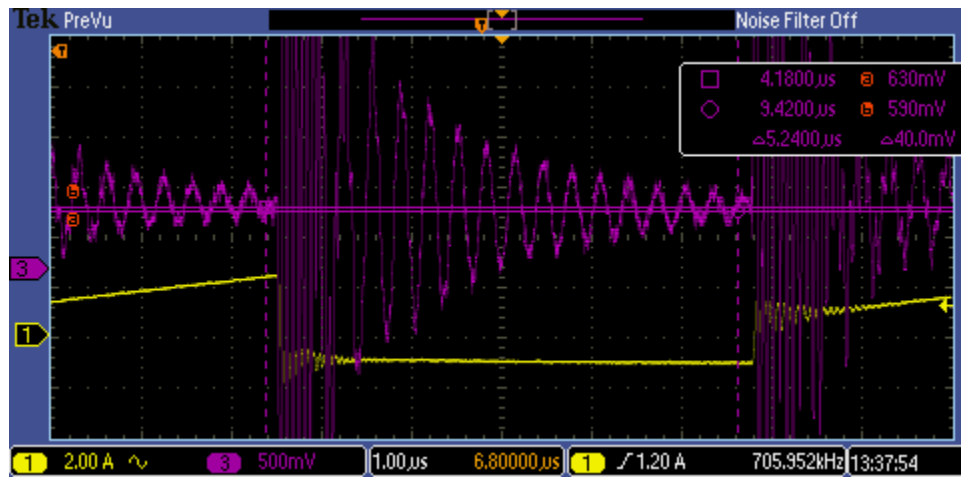


Figure 2.1.1 - 1: Capacitor C1 ESR Voltage Measurement - 500mV and 1us per division

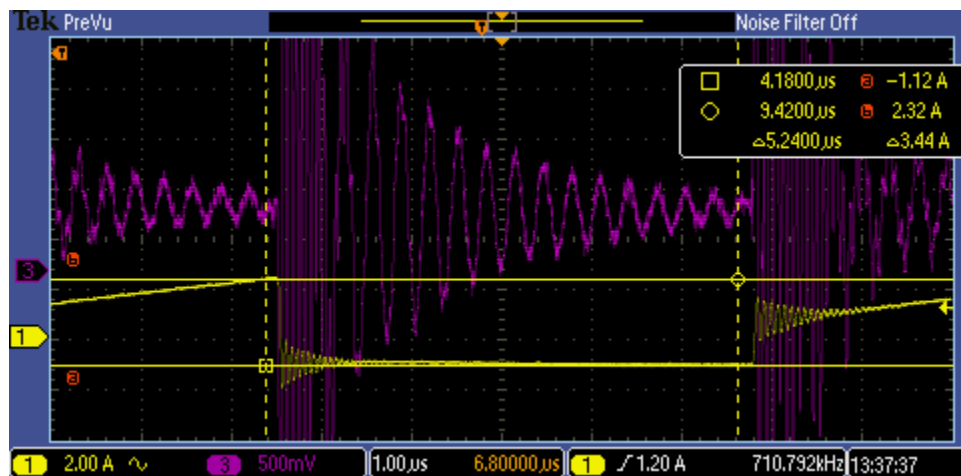


Figure 2.1.1 - 2: Capacitor C1 ESR Current Measurement - 2A and 1us per division

ESR measurements of capacitors are measured using the following formula.

$$ESR = \frac{\Delta V}{\Delta I} = \frac{40mV}{3.44A} = 11.6m\Omega$$

The ESR in the datasheet for this capacitor is listed as 13mΩ. With our measurements, we estimated that the capacitor on the buck converter has an ESR of 11.6mΩ. This is a decent value and around what we expected. The ringing in the current and voltage plots is due to the inductance of the wires on our board. They were necessary in order to obtain scope captures of our circuit elements. This ringing can be reduced by removing our wires and replacing them with shorter leads.

2.1.2 Estimating Power Loss

The capacitor ESR was measured at 11.6mΩ, and from this, a calculation on its power loss can be made. The following formula shows the estimated power loss.

$$Power\ Loss = I_{rms}^2 \times ESR = 4.22^2 \times 0.0116 = 206.6mW$$

The Irms value used in the above calculation came from the capacitor's datasheet. Using the value of 1.32 Irms, which was measured in the laboratory, the power loss then becomes 20.2mW. While any power loss is not ideal, keeping the power loss in the milliwatt range is certainly better than the Watt range.

2.2 Buck Converter Load Testing

2.2.1 15 W Testing

The goal of this test was to create an output power of 15W at 13V, corresponding to 1.15A, by adjusting the converter's duty cycle and load resistance. In order to achieve this, the load resistance on the rheostat was adjusted to 11.4Ω. The duty cycle was adjusted to 73.3% to meet the desired output voltage. The total input power was measured at 17.2V and 920mA to give 15.82W. The total output power was measured as 13V and 1.16A corresponding to 15.08W. This gave the converter an efficiency of 95.3%. Below are plots of the transistor current and the diode voltage.

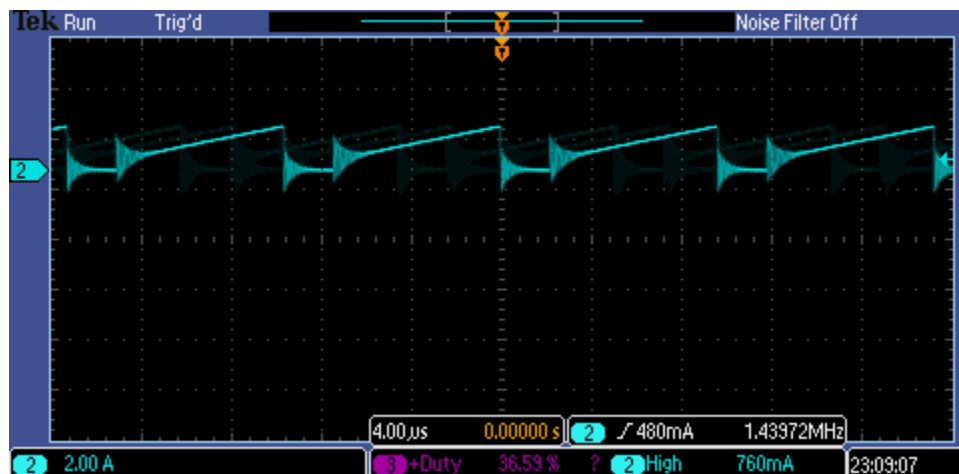


Figure 2.2.1 - 1: Transistor Current

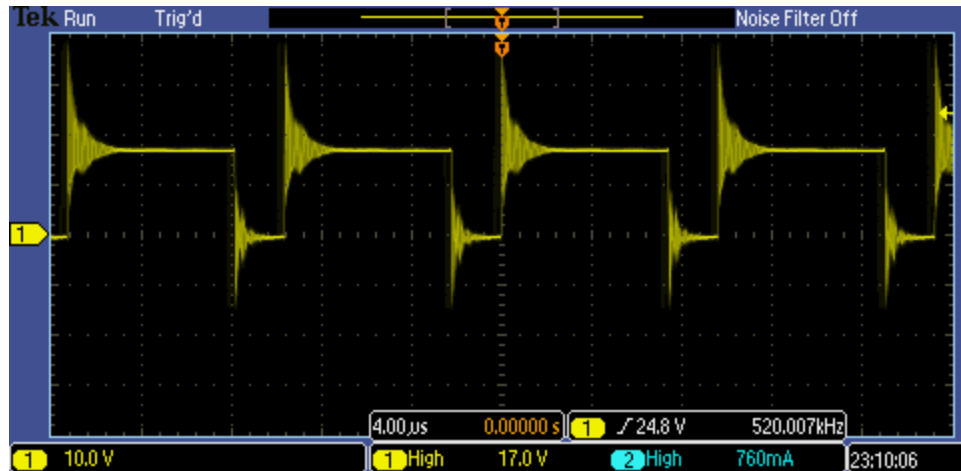


Figure 2.2.1 - 2: Diode Voltage

While the above plots have a large amount of ringing, the converter was still able to operate at a high efficiency. The ringing in these plots can be attributed to loop inductance from long leads between components. While these long leads were necessary at the time, they will be removed seeing that all measurements and scope captures have been taken. Removing these leads will produce less ringing throughout the circuit and provide less losses within the converter.

2.2.2 1 W Testing

The goal of this test was to create an output power of 1.0W at 13V, corresponding to 76.9mA, by adjusting the converter's duty cycle and load resistance. In order to achieve this, the load resistance was adjusted to 165 Ω which allowed for a lower current. The duty cycle then had to be dropped to 23% to meet the desired output voltage. During testing, an output voltage of 12.92V was measured at 78.5mA corresponding to an output power of 1.01W. Our input voltage was 17.2V at 120mA corresponding to an input power of 2.06W. The converter's efficiency was 48.9%. This drop in efficiency can be attributed to the lower duty cycle. Lower duty cycles in buck converters create more losses and should not be run this low. Below are captures of the transistor current and the diode voltage.

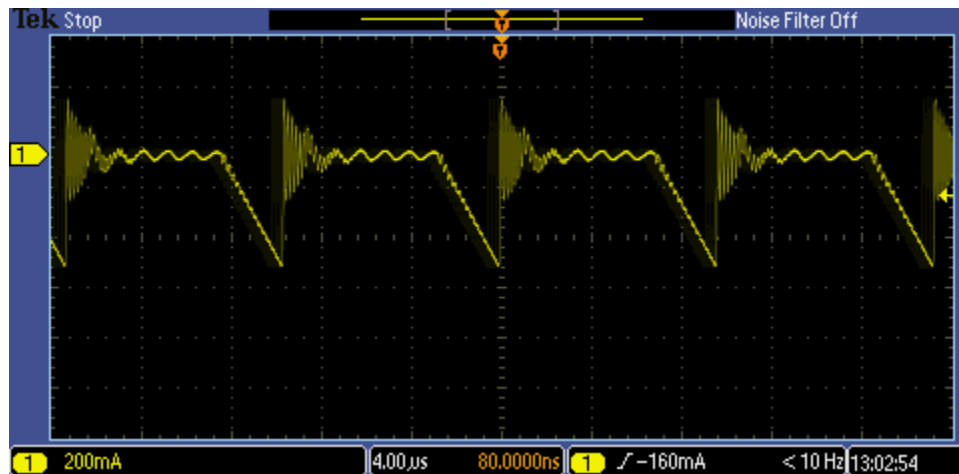


Figure 2.2.2 - 1: Transistor Current

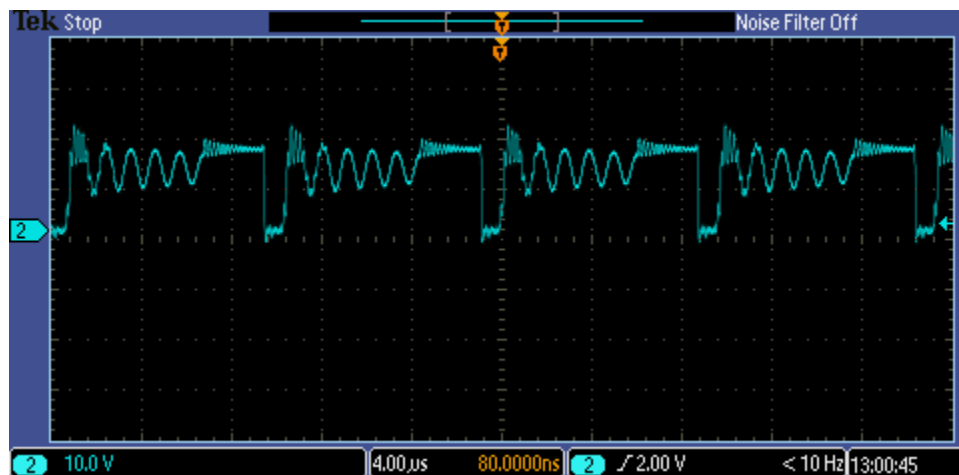


Figure 2.2.2 - 2: Diode Voltage